Authors' Responses

Journal Title:	IEEE Robotics and Automation Letters
Reference Number:	26-3841v1
Title of Paper:	VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuomotor Skill Distillation
Authors:	Anonymous Authors

We would like to extend our heartfelt gratitude to the Editor-in-Chief and the Associate Editor for their time and effort in handling our paper. We also sincerely thank the two anonymous reviewers for their invaluable comments and suggestions, which have greatly helped us improve the quality of our work.

We have completed a major revision of the manuscript and expanded its technical explanations in the following four main areas:

1) validated the necessity and contribution of VLM guidance by adding two ablations that isolate its visual and semantic components and by extending comparisons with fixed and adaptive intervention strategies;
2) improved reproducibility by providing the exact VLM prompt, model and inference settings, visual-input selection procedure, output processing, update frequency, resource usage, failure handling, and relevant training details;
3) strengthened the evaluation of the proposed Bellman success-value critic by reporting task-outcome prediction accuracy, unsafe-event detection, and false-negative rates under student-controlled rollouts; and
4) expanded the real-world evaluation by increasing the number of trials, reporting results for all 22 objects and pooled seen/unseen confidence intervals, adding a physical baseline comparison, and clarifying the empirical scope and limitations of our safety and deployment claims.

The detailed responses to the reviewers' comments are attached. We would be happy to incorporate any further feedback or suggestions into the final version of the manuscript.

RESPONSE TO REVIEWER #14

We sincerely thank the reviewer for the careful assessment of our work and for the constructive suggestions. Below are our point-by-point responses.

Comment 1: ***whether the VLM and visual information provide a clear advantage over simpler adaptive approaches.***

[Response:] We have added ablations (g) and (h) to isolate the contribution of the VLM to the adaptive intervention pipeline. We have also expanded the discussion of Table III in the Results section.

Table 1: Performance of ablations (a)–(h) on four representative objects from the 40-object training set (200 evaluation episodes per object). Each ablation reports lift success and average unsafe rate.

Metric (%)	WindCart		Boot		ToolBox		Cow		**Mean**	
	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe
(a)	75.0	24.0	76.5	28.0	72.0	26.0	70.0	27.5	72.7	26.5
(b)	88.5	13.5	89.0	20.5	82.5	19.0	88.0	17.5	87.1	17.4
(c)	88.0	12.0	89.5	19.5	81.5	18.5	87.0	16.0	86.3	16.7
(d)	74.5	23.0	75.0	27.5	70.5	26.5	69.5	27.0	72.3	25.8
(e)	88.0	7.5	90.0	15.0	82.0	15.5	88.0	12.0	86.8	11.9
(f)	88.5	6.0	90.0	10.5	82.5	11.0	88.5	8.5	87.5	9.3
(g)	88.0	7.5	89.5	12.5	82.0	13.0	88.0	10.5	86.9	10.2
(h)	86.5	7.0	88.5	14.0	81.0	13.5	87.0	11.0	85.7	11.7

The results support our claim that VLM-based intervention outperforms both LLM-based intervention and performance-based curriculum adaptation. Semantic information alone reduces the unsafe exploration rate, while combining visual and semantic information enables more precise intervention, achieving the highest task success rate and the lowest unsafe exploration rate. Although the performance-based curriculum reduces unsafe exploration to some extent, it sacrifices task success, suggesting that a simple rule-based method is insufficient for guiding intervention policies in complex scenarios. The detailed design of the performance-curriculum ablation (h) is provided in Supplementary Section C.5.

Comment 2: ***The related work on adaptive intervention should also be expanded.***

[Response:] We have reformulated the Related Work section to more clearly discuss adaptive intervention methods. In particular, AIM provides an important foundation for our work, and we have added it to the reference list. Compared with prior methods such as HG-DAgger and ThriftyDAgger, our proposed pipeline does not require human intervention, enabling fully autonomous intervention during online policy distillation.

Adaptive Intervention in Interactive Imitation Learning. Many imitation learning methods ~~operate offline, using a fixed dataset of expert demonstrations to match the expert action distribution~~use fixed offline datasets for supervision [14], [15]. In sequential decision-making, however, ~~distributional~~ distribution shift induces covariate shift and compounding errors. Interactive imitation learning ~~[16] mitigates this issue by collecting supervision online.~~ [16] mitigates these issues by collecting online supervision along trajectories visited by the learner.

~~DAgger~~DAgger [10] addresses covariate shift by iteratively ~~rolling out the learner and relabeling visited~~ relabeling learner-visited states with an expert, ~~thereby aggregating data under the learner-induced state distribution. SafeDAgger [12] extends DAgger with a teacher-intervened rollout policy that takes over~~ while SafeDAgger [12] adds teacher intervention in high-risk states to reduce unsafe exploration. ~~Recent variants [17], [18] extend interactive imitation learning to robotic manipulation, yet often rely on human supervision~~

Later methods introduce adaptive intervention mechanisms based on uncertainty-driven rules [19], human gating [20]

Comment 3: ***the authors should provide the VLM prompt and explain how the images are selected and how the VLM outputs are used.***

[Response:] The details required to reproduce our method are provided in the supplementary material. The exact VLM prompt and representative visual input examples are included in Supplementary Section B.

We used OpenAI GPT-5.5 with the rollout statistics above and up to eight high-detail RGB images encoded as JPEG data URLs. We queried the model after 2,000 episodes and every 10,000 episodes thereafter (10 calls over 100,000 episodes). Archived reports from the earlier low-detail setting contain 30,641 prompt and 4,841 completion tokens (35,482 total).

The requested machine-readable output is a top-level JSON object with the following schema:

```
{
  "delta_l2_threshold": <number>, "delta_success_threshold": <number>,
  "confidence": <number>, "reason": <string>
}
```

The two returned values are threshold displacements. For each channel $x \in \{\delta, Q\}$, the candidate threshold is

$$\tilde{\tau}_x^{(k)} = \tau_x^{(k)} + \Delta\tau_x^{(k)}.$$

Each candidate is clipped to a fixed range around its initial value and then exponentially smoothed with the previously applied threshold:

$$\bar{\tau}_x^{(k)} = \mathrm{clip}\left(\tilde{\tau}_x^{(k)}, c_x^{\min}\tau_x^{(0)}, c_x^{\max}\tau_x^{(0)}\right), \qquad \tau_x^{(k+1)} = (1-\alpha)\tau_x^{(k)} + \alpha\bar{\tau}_x^{(k)}.$$

We use $\alpha = 0.1$ and $(c_x^{\min}, c_x^{\max}) = (0.5, 1.5)$ for both thresholds. The updated values are applied from the next control step through the intervention gate $\|a_t^S - a_t^T\|_2 > \tau_\delta \ \vee \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q$; lowering τ_δ or raising τ_Q increases teacher intervention. The `confidence` and `reason` fields are logged for diagnostics and do not alter the update. This adaptation is used only during simulation distillation; physical deployment uses the frozen RGB student alone.

If a VLM request fails or its output cannot be parsed, the controller retains the previously applied thresholds and tries again at the next scheduled update. None of the reported trials encountered a condition that activated this fallback.

The algorithm block gives the full prompt template used by the runtime implementation. Braced terms denote values serialized at query time, and the visual-sample block is repeated once for each attached image.

VLM Prompt Example

```
System message:
You tune safety-threshold displacements and return
compact JSON only.
User message:
You are a conservative threshold advisor for
DexSafeDagger robot distillation.
Do not decide intervention directly.  Recommend
threshold displacements for the existing gates.
Task:  Propose displacements for the
teacher-student disagreement and predicted-success
thresholds used in dexterous robot distillation.
Existing arbitration:
- intervene if teacher_student_l2 >= l2_threshold
- or if predictor_success <= success_threshold
Return only strict JSON with keys:
delta_l2_threshold, delta_success_threshold,
confidence, reason.
Keep changes conservative.  Prefer small threshold
shifts unless the recent window shows clear over-
or under-intervention.
Base thresholds and clamps:
- base_l2_threshold={base_l2_threshold}
- allowed_l2_range=[{l2_min}, {l2_max}]
- base_success_threshold={base_success_threshold}
- allowed_success_range=[{success_min},
{success_max}]
Current advisor thresholds:
- current_l2_threshold={current_l2_threshold}
- current_success_threshold={current_success_threshold}
Recent rollout statistics:
{
 "frame":  {frame}, "intervention_rate":  {beta},
"step":  {step},
 "sample_count":  {N}, "l2_threshold":
{tau_delta},
 "success_threshold":  {tau_Q},
"unsafe_episode_rate":  {r_unsafe},
 "l2":  {"min":  {v}, "max":  {v}, "mean":  {v},
     "p10":  {v}, "p50":  {v}, "p90":  {v}},
 "success":  {"min":  {v}, "max":  {v}, "mean":
{v},
     "p10":  {v}, "p50":  {v}, "p90":  {v}},
 "unsafe_reason_prop":  { "hand_too_far":  {v},
     "harmful_collision":  {v},
"object_out_of_bound":  {v},
     "palm_flipped":  {v}},
 "visual_buffer_size":  {B}, "visual_samples_attached":
{M},
 "visual_samples":  [{ "env_id":  {v}, "frame":
{v},
     "image_attached":  true, "image_key":
{v}, "source":  {high_l2 | low_success |
unsafe_triggered},
     "l2_threshold":  {v}, "success_threshold":
{v},
     "teacher_student_l2":  {v},
"predictor_success":  {v},
     "object_id":  {v}, "object_name":  {v},
"object_env_count":  {v},
     "step":  {v}, "unsafe":  {true | false}}, ...]

}
For each selected visual sample i = 1,...,M:
Visual sample {i} metadata:  {the corresponding
record above, excluding "image_attached"}
<IMAGE {i}:  JPEG data URL, detail="high">
```

Comment 4: ***The Bellman success-value critic is also important for intervention, but its performance is not evaluated in detail. Some evaluation of the critic, especially its errors for unsafe cases, would be helpful.***

[Response:] We use the resulting student and fixed-threshold Bellman critic from ablation (e) to evaluate the critic's performance. Both networks are frozen during evaluation, and the critic runs in parallel as a passive evaluator. We evaluate 200 episodes for each of the 40 training objects in simulation. Table 2 reports the observed lift-success and unsafe-episode rates together with the task-outcome prediction accuracy, the

unsafe-episode detection rate, and its corresponding false-negative rate.

The lift-success rate and unsafe-episode rate characterize the student policy's ability to accomplish the task and the frequency of unsafe outcomes, respectively. Task-outcome prediction accuracy is the percentage of episodes for which the critic correctly predicts the task outcome. The unsafe-episode detection rate is the percentage of unsafely terminated episodes that are flagged by the critic before termination. The unsafe false-negative rate is the complementary percentage of unsafe episodes that the critic does not flag before termination.

Table 2: Evaluation of the frozen student and fixed-threshold Bellman critic obtained from ablation (e) under student-controlled rollouts.

Metric	Result
Lift-success rate	86.8
Unsafe-episode rate	11.9
Task-outcome prediction accuracy	90.3
Unsafe-episode detection rate	94.2
Unsafe false-negative rate	5.8

Comment 5: ***Please also clarify the number of runs/random seeds used for the results.***

[Response:] We run 30 real-world evaluations for each object, distributed equally across random seeds 40, 41, 42, 43, and 44 (six evaluations per seed).

Comment 6: ***Finally, I am not fully convinced by the use of the term "safety-critical." The current safety evaluation is based on task-specific unsafe conditions rather than formal safety guarantees. The authors should clarify the scope of this claim.***

[Response:] We acknowledge that the term "safety-critical" is too strong for an adaptive intervention pipeline such as ours. We have therefore revised the title to "VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuomotor Skill Distillation" and removed or qualified the term elsewhere in the manuscript. The revised text explicitly states that our method provides empirical risk reduction under task-specific event definitions rather than a formal safety guarantee, and that safety violations remain possible.

Comment 7: ***However, only VLM-DexSafeDagger evaluated on the real robot does not sufficiently prove the proposed concept.***

[Response:] We deploy the frozen RGB student policies resulting from ablations (e) and (f) on the real robot using the same object set, initial-condition protocol, and number of trials. We report lift-success and unsafe-event rates for the seen and unseen object sets in Main Paper Table IV.

Table 3: Overall real-world comparison of the frozen student policies resulting from ablations (e) and (f). Rates are reported separately for the 14 seen and 8 unseen objects.

Metric (%)	Ablation (e)		Ablation (f)	
	Seen (14)	Unseen (8)	Seen (14)	Unseen (8)
Lift success	83.4	61.7	83.6	64.6
Unsafe-event rate	16.1	41.4	14.8	39.2

Only the resulting student policy is deployed; the teacher, Bellman critic, VLM, and threshold-adaptation mechanism are not active during physical evaluation.

Comment 8: ***It would be helpful to provide overall results for all 22 objects, for example by reporting results separately for seen and unseen objects.***

[Response:] We have doubled the number of evaluation episodes for each object and reported the per-object and aggregate seen/unseen results for all 22 objects in Supplementary Tables IV and V.

Table 4: Per-object results for the remaining 16 objects (30 episodes per object). **OOD objects** are shaded.

Metric (%)	Horse	Teapot	Sphinx	Dragon	Homer	Ogre	Chebura shka	Lucy
Lift Success	83.3	90.0	80.0	86.7	76.7	83.3	90.0	80.0
Unsafe Rate Avg	16.7	6.7	13.3	13.3	23.3	10.0	6.7	16.7
Harmful collision	6.7	3.3	3.3	3.3	3.3	3.3	3.3	3.3
Object escape	6.7	3.3	6.7	10.0	10.0	6.7	3.3	10.0
Palm flipped	3.3	0.0	3.3	0.0	6.7	0.0	0.0	3.3
Hand too far	0.0	0.0	0.0	0.0	3.3	0.0	0.0	0.0

Metric (%)	Beethoven	Igea	**Dog Pair**	**Cow statue**	**Buddha statue**	**Blub**	**Monkey head**	**Dinosaur**
Lift Success	86.7	73.3	**70.0**	**63.3**	**73.3**	**60.0**	**66.7**	**56.7**
Unsafe Rate Avg	6.7	13.3	**33.3**	**40.0**	**33.3**	**43.3**	**36.7**	**46.7**
Harmful collision	0.0	3.3	**13.3**	**10.0**	**13.3**	**23.3**	**16.7**	**26.7**
Object escape	6.7	3.3	**13.3**	**16.7**	**13.3**	**16.7**	**16.7**	**16.7**
Palm flipped	0.0	3.3	**3.3**	**10.0**	**3.3**	**3.3**	**3.3**	**3.3**
Hand too far	0.0	3.3	**3.3**	**3.3**	**3.3**	**0.0**	**0.0**	**0.0**

Table 5: Aggregate real-world performance across 14 seen and 8 unseen objects. Mean rates are pooled over 30 episodes per object (420 seen and 240 unseen episodes); parentheses report Wilson 95% confidence intervals.

Metric (%)	Seen mean (95% CI)	Unseen mean (95% CI)
Lift success	83.6 (79.7–86.8)	64.6 (58.3–70.4)
Unsafe rate avg.	14.7 (11.7–18.5)	39.2 (33.2–45.5)
Harmful collision	4.7 (3.1–7.2)	18.8 (14.3–24.2)
Object escape	8.1 (5.9–11.1)	15.8 (11.8–21.0)
Palm flipped	1.4 (0.7–3.1)	3.3 (1.7–6.4)
Hand too far	0.5 (0.1–1.7)	1.3 (0.4–3.6)

As recommended, we have also extended Tables II–IV in the main paper to report aggregate results for each

evaluation setting.

RESPONSE TO REVIEWER #22

We sincerely thank the reviewer for the detailed and constructive comments. Below are our point-by-point responses.

Comment 1: ***The term "safety-critical" is too strong because the framework supplies empirical intervention rather than formal constraint satisfaction. State explicitly that safety violations remain possible.***

[Response:] We acknowledge that the term "safety-critical" is too strong for an adaptive intervention pipeline such as ours. We have therefore revised the title to "VLM-DexSafeDagger: VLM-Guided Intervention Policy Towards Safer Visuomotor Skill Distillation" and removed or qualified the term elsewhere in the manuscript. The revised text explicitly states that our method provides empirical risk reduction under task-specific event definitions rather than a formal safety guarantee, and that safety violations remain possible.

Comment 2: ***Demonstrate that the VLM is necessary by comparing against fixed thresholds, rule-based adaptation, Bayesian optimization, curriculum schedules, grid search, and a small learned threshold network.***

[Response:] We have supplemented the ablation study with variants (g) and (h). Together with variants (e) and (f), these experiments form a controlled ladder of adaptive intervention strategies: fixed thresholds, a performance-based curriculum, semantics-guided intervention-policy adaptation, and visual-semantic-guided adaptive intervention. We have added a corresponding discussion to the Results section. Table III shows that adaptive intervention reduces the unsafe rate relative to fixed thresholds. However, the curriculum-based method sacrifices lift success, whereas incorporating semantic information recovers task performance while further reducing unsafe exploration. Overall, combining visual and semantic information outperforms the other approaches, achieving the best balance between lift success and unsafe rate.

The detailed design of the performance-curriculum ablation (h) is provided in Supplementary Section C.5.

Table 6: Performance of ablations (a)–(h) on four representative objects from the 40-object training set (200 evaluation episodes per object). Each ablation reports lift success and average unsafe rate.

Metric (%)	WindCart		Boot		ToolBox		Cow		**Mean**	
	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe	Lift	Unsafe
(a)	75.0	24.0	76.5	28.0	72.0	26.0	70.0	27.5	72.7	26.5
(b)	88.5	13.5	89.0	20.5	82.5	19.0	88.0	17.5	87.1	17.4
(c)	88.0	12.0	89.5	19.5	81.5	18.5	87.0	16.0	86.3	16.7
(d)	74.5	23.0	75.0	27.5	70.5	26.5	69.5	27.0	72.3	25.8
(e)	88.0	7.5	90.0	15.0	82.0	15.5	88.0	12.0	86.8	11.9
(f)	88.5	6.0	90.0	10.5	82.5	11.0	88.5	8.5	87.5	9.3
(g)	88.0	7.5	89.5	12.5	82.0	13.0	88.0	10.5	86.9	10.2
(h)	86.5	7.0	88.5	14.0	81.0	13.5	87.0	11.0	85.7	11.7

Comment 3: ***The Bellman critic predicts combined "safe-lift-success," thereby conflating task success***

with safety. Separate safety risk and task value, and report calibration, false-negative rates, and reliability diagrams.

[Response:] Lift success and unsafe termination measure different aspects of an episode and are not mutually exclusive. Lift success is recorded once the object satisfies the prescribed height-and-hold criterion, but satisfying this criterion does not terminate the rollout. The episode continues to be monitored for unsafe events until timeout or unsafe termination. Consequently, an episode may achieve lift success and subsequently terminate because of an unsafe event. The Bellman critic is therefore trained on the stricter joint safe-lift-success target, which is positive only when the lift criterion is satisfied without an unsafe event.

We evaluate the resulting student and fixed-threshold Bellman critic from ablation (e). Both networks are frozen, teacher intervention is disabled, and the critic runs in parallel as a passive evaluator. We separately report task success and unsafe termination outcomes, including the critic's unsafe-episode detection and false-negative rates. The results are reported in Supplementary Section C.1.

Table 7: Evaluation of the frozen student and fixed-threshold Bellman critic obtained from ablation (e) under student-controlled rollouts.

Metric	Result
Lift-success rate	86.8
Unsafe-episode rate	11.9
Task-outcome prediction accuracy	90.3
Unsafe-episode detection rate	94.2
Unsafe false-negative rate	5.8

Comment 4: ***Teacher intervention is assumed safe, yet teacher policies may collide or fail. Quantify teacher unsafe rates and define how the framework responds when teacher and critic disagree.***

[Response:] Teacher takeover is not guaranteed to be safe; it provides an empirical tendency toward safer exploration rather than a formal guarantee. We report the teacher policies' unsafe rates in Table III of the main paper and, in this revision, provide mean statistics across all tested objects. The intervention gate follows a conservative OR rule: intervention is triggered if either the teacher–student action disagreement exceeds its threshold or the critic's predicted success falls below its threshold. Thus, when the two signals disagree, the conservative outcome is to intervene. Comparing ablations (c) and (e) shows that adding the critic-based branch substantially reduces the unsafe exploration rate.

Comment 5: ***Provide exact VLM prompts, model version, decoding settings, image-selection procedure, output parsing, update frequency, latency, cost, and failure handling. Reproducibility may otherwise depend on an external proprietary service.***

[Response:] The details required to reproduce our method are provided in the supplementary material. The exact VLM prompt, representative visual-input examples, model and inference configuration, image-selection and output-processing procedures, update frequency, token usage, and failure handling are included in Supplementary Section B.

The algorithm block gives the full prompt template used by the runtime implementation. Braced terms denote values serialized at query time, and the visual-sample block is repeated once for each attached image.

VLM Prompt Example

```
System message:
You tune safety-threshold displacements and return
compact JSON only.
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You are a conservative threshold advisor for
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Do not decide intervention directly.  Recommend
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Task:  Propose displacements for the
teacher-student disagreement and predicted-success
thresholds used in dexterous robot distillation.
Existing arbitration:
- intervene if teacher_student_l2 >= l2_threshold
- or if predictor_success <= success_threshold
Return only strict JSON with keys:
delta_l2_threshold, delta_success_threshold,
confidence, reason.
Keep changes conservative.  Prefer small threshold
shifts unless the recent window shows clear over-
or under-intervention.
Base thresholds and clamps:
- base_l2_threshold={base_l2_threshold}
- allowed_l2_range=[{l2_min}, {l2_max}]
- base_success_threshold={base_success_threshold}
- allowed_success_range=[{success_min},
{success_max}]
Current advisor thresholds:
- current_l2_threshold={current_l2_threshold}
- current_success_threshold={current_success_threshold}
```

```
Recent rollout statistics:
{
 "frame":  {frame}, "intervention_rate":  {beta},
"step":  {step},
 "sample_count":  {N}, "l2_threshold":
{tau_delta},
 "success_threshold":  {tau_Q},
"unsafe_episode_rate":  {r_unsafe},
 "l2":  {"min":  {v}, "max":  {v}, "mean":  {v},
     "p10":  {v}, "p50":  {v}, "p90":  {v}},
 "success":  {"min":  {v}, "max":  {v}, "mean":
{v},
     "p10":  {v}, "p50":  {v}, "p90":  {v}},
 "unsafe_reason_prop":  { "hand_too_far":  {v},
     "harmful_collision":  {v},
"object_out_of_bound":  {v},
     "palm_flipped":  {v}},
 "visual_buffer_size":  {B}, "visual_samples_attached":
{M},
 "visual_samples":  [{ "env_id":  {v}, "frame":
{v},
     "image_attached":  true, "image_key":
{v}, "source":  {high_l2 | low_success |
unsafe_triggered},
     "l2_threshold":  {v}, "success_threshold":
{v},
     "teacher_student_l2":  {v},
 "predictor_success":  {v},
     "object_id":  {v}, "object_name":  {v},
"object_env_count":  {v},
     "step":  {v}, "unsafe":  {true | false}}, ...]

}
For each selected visual sample i = 1,...,M:
Visual sample {i} metadata:  {the corresponding
record above, excluding "image_attached"}
<IMAGE {i}:  JPEG data URL, detail="high">
```

We used OpenAI GPT-5.5 with the rollout statistics above and up to eight high-detail RGB images encoded as JPEG data URLs. We queried the model after 2,000 episodes and every 10,000 episodes thereafter (10 calls over 100,000 episodes). Archived reports from the earlier low-detail setting contain 30,641 prompt and 4,841 completion tokens (35,482 total).

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The two returned values are threshold displacements. For each channel $x \in \{\delta, Q\}$, the candidate threshold is

$$\tilde{\tau}_x^{(k)} = \tau_x^{(k)} + \Delta\tau_x^{(k)}.$$

Each candidate is clipped to a fixed range around its initial value and then exponentially smoothed with the previously applied threshold:

$$\bar{\tau}_x^{(k)} = \mathrm{clip}\left(\tilde{\tau}_x^{(k)}, c_x^{\min}\tau_x^{(0)}, c_x^{\max}\tau_x^{(0)}\right), \qquad \tau_x^{(k+1)} = (1-\alpha)\tau_x^{(k)} + \alpha\bar{\tau}_x^{(k)}.$$

We use $\alpha = 0.1$ and $(c_x^{\min}, c_x^{\max}) = (0.5, 1.5)$ for both thresholds. The updated values are applied from the next control step through the intervention gate $\|a_t^S - a_t^T\|_2 > \tau_\delta \ \vee \hat{Q}_{\text{succ}}(s_t, a_t^S) < \tau_Q$; lowering τ_δ or raising τ_Q increases teacher intervention. The `confidence` and `reason` fields are logged for diagnostics and do not alter the update. This adaptation is used only during simulation distillation; physical deployment uses the frozen RGB student alone.

If a VLM request fails or its output cannot be parsed, the controller retains the previously applied thresholds and tries again at the next scheduled update. None of the reported trials encountered a condition that activated this fallback.

Comment 6: ***Real-world unseen-object performance is weak: lift success falls to 60–66.7%, while unsafe episodes reach 40%. Claims of sufficient deployment robustness should be moderated.***

[Response:] We will remove the claim that the current results establish sufficient robustness for deployment.

Comment 7: ***Only fifteen real-world episodes per reported object produce coarse estimates. Increase repetitions and report confidence intervals and statistical comparisons across all 22 objects.***

[Response:] We have doubled the number of real-world evaluation episodes from 15 to 30 per object. We have also reported the complete results for all 22 tested objects in Supplementary Section C.3. Supplementary Table V reports the pooled means and Wilson 95% confidence intervals for the 14 seen and eight unseen objects.

Table 8: Per-object results for the remaining 16 objects (30 episodes per object). **OOD objects** are shaded.

Metric (%)	Horse	Teapot	Sphinx	Dragon	Homer	Ogre	Chebura shka	Lucy
Lift Success	83.3	90.0	80.0	86.7	76.7	83.3	90.0	80.0
Unsafe Rate Avg	16.7	6.7	13.3	13.3	23.3	10.0	6.7	16.7
Harmful collision	6.7	3.3	3.3	3.3	3.3	3.3	3.3	3.3
Object escape	6.7	3.3	6.7	10.0	10.0	6.7	3.3	10.0
Palm flipped	3.3	0.0	3.3	0.0	6.7	0.0	0.0	3.3
Hand too far	0.0	0.0	0.0	0.0	3.3	0.0	0.0	0.0

Metric (%)	Beethoven	Igea	**Dog Pair**	**Cow statue**	**Buddha statue**	**Blub**	**Monkey head**	**Dinosaur**
Lift Success	86.7	73.3	**70.0**	**63.3**	**73.3**	**60.0**	**66.7**	**56.7**
Unsafe Rate Avg	6.7	13.3	**33.3**	**40.0**	**33.3**	**43.3**	**36.7**	**46.7**
Harmful collision	0.0	3.3	**13.3**	**10.0**	**13.3**	**23.3**	**16.7**	**26.7**
Object escape	6.7	3.3	**13.3**	**16.7**	**13.3**	**16.7**	**16.7**	**16.7**
Palm flipped	0.0	3.3	**3.3**	**10.0**	**3.3**	**3.3**	**3.3**	**3.3**
Hand too far	0.0	3.3	**3.3**	**3.3**	**3.3**	**0.0**	**0.0**	**0.0**

Table 9: Aggregate real-world performance across 14 seen and 8 unseen objects. Mean rates are pooled over 30 episodes per object (420 seen and 240 unseen episodes); parentheses report Wilson 95% confidence intervals.

Metric (%)	Seen mean (95% CI)	Unseen mean (95% CI)
Lift success	83.6 (79.7–86.8)	64.6 (58.3–70.4)
Unsafe rate avg.	14.7 (11.7–18.5)	39.2 (33.2–45.5)
Harmful collision	4.7 (3.1–7.2)	18.8 (14.3–24.2)
Object escape	8.1 (5.9–11.1)	15.8 (11.8–21.0)
Palm flipped	1.4 (0.7–3.1)	3.3 (1.7–6.4)
Hand too far	0.5 (0.1–1.7)	1.3 (0.4–3.6)

Comment 8: ***Safety-event thresholds for harmful collision, object escape, palm flipping, and workspace violation require physical justification. Include contact-force measurements and severity-weighted safety metrics.***

[Response:] We have provided the exact definitions of all unsafe events in the termination-conditions discussion in Supplementary Section A.2. When multiple unsafe conditions occur simultaneously during training, the event label is assigned according to a clear severity-based priority: harmful collision, object escape, palm flipped, and hand too far, in descending order.

Termination Conditions. We define four unsafe conditions that serve both as termination criteria during teacher-policy training and as unsafe-event signals during student-policy distillation. We additionally terminate an episode when it reaches the maximum time horizon; however, a timeout is not classified as an unsafe event. All five termination conditions are checked at 60 Hz in the environment frame.

(a) *Harmful collision:* We monitor all arm–table contacts and trigger this termination when the palm or any fingertip is less than 1 cm above the tabletop, or when any arm link contacts the table. An arm–table contact is registered when the ℓ_1 norm of the filtered contact-force vector exceeds 0.1 N.

(b) *Object escape:* We define a permissible object workspace above the table that covers the intended manipulation region. We trigger object-escape termination as soon as the object center leaves this space or falls below the table.

(c) *Palm flipped:* The palm should face downward toward the object and table. We trigger this termination when its downward direction differs from world-down by more than 90°, at which point the nominal grasp geometry is inverted.

(d) *Hand too far:* We use the same permissible workspace as in (b) and trigger this termination when the palm link leaves it, indicating that the hand has moved too far during exploration.

(e) *Timeout:* We terminate the episode when the time index reaches the maximum episode horizon, $t \geq T_{\max}$.

Expanding the termination rule used in the main paper gives

$$\mathbb{I}_{\text{term}}(s_t, t) = \mathbb{I}_{\text{coll}}(s_t) \vee \mathbb{I}_{\text{obj}}(s_t) \vee \mathbb{I}_{\text{flip}}(s_t) \vee \mathbb{I}_{\text{hf}}(s_t) \vee \mathbb{I}_{\text{timeout}}.$$

To avoid conflicts between simultaneous termination conditions, we assign the event label using the following descending severity priority: harmful collision $\succ$ object escape $\succ$ palm flipped $\succ$ hand too far $\succ$ timeout.

This order reflects decreasing immediacy of harm and loss of control: direct arm–table contact poses the clearest hardware risk, object escape represents loss of containment, palm inversion and workspace departure

are geometric failures of decreasing immediate consequence, and timeout is last because it is not classified as unsafe.

Comment 9: ***Clarify whether VLM adaptation occurs only during simulation distillation or during physical deployment. If absent at deployment, the role of scene-level VLM reasoning should not be overstated.***

[Response:] The VLM is queried only during simulation-based online distillation. During physical deployment, the RGB student acts without the VLM, privileged teacher, Bellman critic, or threshold adaptation. We will state this distinction in the abstract, method, experimental setup, and limitations, and will describe the VLM as a training-time adaptation mechanism rather than a deployment-time safety monitor.

Comment 10: ***Compare training samples, teacher queries, interventions, computation, and final success across Vanilla DAgger, SafeDAgger, critic-based intervention, and VLM-guided intervention using multiple random seeds.***

[Response:] We repeat the complete distillation process for Vanilla DAgger (b), Vanilla SafeDAgger (c), critic-based intervention (e), and VLM-guided intervention (f) using five independent random seeds, $\{40, 41, 42, 43, 44\}$. All variants use the same number of training samples, and the teacher-intervention rates exhibit only minor variation across seeds. The final lift-success and unsafe-episode rates are reported as the mean $\pm$ standard deviation across seeds in Supplementary Section C.2.

Table 10: Comparison of Vanilla DAgger, Vanilla SafeDAgger, critic-based intervention, and VLM-guided intervention across five random seeds $\{40, 41, 42, 43, 44\}$. Lift success and unsafe rate are reported as mean $\pm$ standard deviation across seeds.

Metric	Vanilla DAgger (b)	Vanilla SafeDAgger (c)	Critic-based intervention (e)	VLM-guided intervention (f)
Lift Success (%) $\uparrow$	87.1 ± 0.99	86.3 ± 0.92	86.8 ± 0.74	87.5 ± 0.98
Unsafe Rate (%) $\downarrow$	17.4 ± 0.52	16.7 ± 0.48	11.9 ± 0.58	9.3 ± 0.39

We thank the Editor and the reviewers again for their thoughtful feedback. The requested revisions will help us present the method's contribution more clearly, support its empirical claims more rigorously, and state its limitations more precisely.